

Building a Stanley Cup Roster: Does Roster Composition Predict NHL Playoff Success?

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1. Introduction

Every spring, the same debate comes up around the NHL: playoff hockey is different. The game is believed to get tighter and more physical in the playoffs, and the common belief is that teams need size and toughness to survive four rounds of best-of-seven series. This belief shapes real decisions, as general managers add big, physical players at the trade deadline and build their rosters with the playoffs in mind. However, it has rarely been tested with data. Chan, Cho, and Novati (2012) found that physical forwards and physical defensemen did not significantly contribute to a team's regular-season success, but whether that changes once the playoffs begin is still an open question.

This project asks whether the makeup of a roster matters for playoff success. More specifically, given how strong a team was during the regular season, is the mix of player types it uses in the playoffs associated with how far it goes? Framing the question this way matters. Better teams naturally go further in the playoffs, so the goal is not to identify what good teams look like, but to find which roster makeups do better or worse in the playoffs than their regular-season record would predict. This is also an important question because NHL playoff results are very unpredictable. Lopez, Matthews, and Baumer (2018) found that the NHL is one of the most random of the major North American leagues, so if roster composition explains anything beyond regular-season strength, that would be valuable information for teams building a roster.

To answer this question, I used NHL data from 21 seasons, from 2005-06 to 2025-26. First, I used k-means clustering to sort players into five forward types and four defense types based on their size, physical play, production, and special teams role. Then, I measured each of the 336 playoff teams' composition by how much of its playoff ice time went to each player type. Finally, I regressed the number of playoff rounds each team won on its composition, while controlling for regular-season points percentage, save percentage, and season. I also checked the results with an ordered logit model and two robustness checks.

The results show that playoff success is largely unpredictable, and roster composition as a whole is not a statistically significant predictor once regular-season strength is accounted for. However, two patterns hold up across every version of the model. Teams that give more of their defense ice time to offensive, puck-moving defensemen tend to go further, while teams that give more ice time to physical forwards tend to have shorter playoff runs. I find no evidence that size or physical play helps teams in the playoffs, which goes against the common belief.

2. Literature Review

The idea of sorting hockey players into types has a long history in sports economics and analytics. One of the earliest examples is Jones, Nadeau, and Walsh (1997), who used a clustering procedure on penalty minutes and points per game to split NHL players from the 1989-90 season into "grunts," the physical players, and skill players. They found that weight was a significant predictor of being a grunt and that grunts and skill players are paid under completely different salary structures, although grunts did not earn more overall. Their work established that the physical player is a real and distinct category on NHL rosters, not just a label

used by fans. Vincent and Eastman (2009), as described by Chan, Cho, and Novati (2012), later applied cluster analysis to players' career statistics up to 2002-03 to define styles of play and compare earnings between them. Both studies focused on how player types are paid, rather than how they affect winning.

The article most closely related to this project is Chan, Cho, and Novati (2012), published in *Interfaces*. Using regular-season data from 2005-06 to 2009-10, they used k-means clustering to define four forward types (Top Line, Second Line, Defensive, and Physical), four defense types (Offensive, Defensive, Average, and Physical), and three goalie types. They then regressed each team's regular-season points on the average number of players of each type it had on the ice, weighted by ice time. Goalies contributed the most to team performance, followed by forwards and then defensemen. Most importantly for this project, Physical forwards and Physical defensemen were not statistically significant contributors to regular-season points. Chan et al. also found that defensive forwards provided the most value relative to their salary, and they used their model to build a tool for evaluating trades. This project borrows much of their design, including the k-means clustering, rate statistics divided by ice time, separate clustering by position, and ice-time-weighted team compositions. However, their study only examines the regular season, and the playoffs are never analyzed or proposed as future work.

More recent work has continued to refine how player roles are defined and has started to connect roles to team success. Säfvenberg, Carlsson, and Lambrix (2024), which cites Chan et al. (2012), used around 30 variables per 60 minutes, including height, weight, fights, and the share of ice time spent on the power play and penalty kill, to identify four forward roles and three defense roles from the 2021-22 season. Instead of placing each player in a single cluster, they used fuzzy clustering, which gives each player a probability of belonging to each role. They then compared the role makeup of each team's 18 most-used skaters to whether it made the playoffs. Playoff teams tended to have more offensive forwards and offensive defensemen, physical forwards showed no clear difference because most teams had very few of them, and physical and defensive defensemen were more common on non-playoff teams. However, their analysis covers one season, only measures whether a team made the playoffs, and does not control for team strength, and the authors themselves suggest extending it to multiple seasons. Outside of hockey, Penner (2025) used k-means to group 22,500 basketball players from 110 leagues into nine archetypes and then derived an "optimal" championship roster from the archetype makeup of top NBA lineups and international medal-winning teams. Successful teams had more scorers and playmakers, while low-efficiency defenders and frequent foulers were rarely used. Notably, Penner left playoff statistics out of the clustering because of their small samples and different style of play. Still, Penner's optimal proportions are descriptive, and they do not account for how good the teams already were.

A separate line of research has looked directly at whether physical play and size lead to winning. Leard and Doyle (2011) studied every game of the 2007-08 season and found that winning fights does not lead to winning games, even after restricting their sample to first-period fights to avoid the problem of losing teams starting fights late in games. They also found that save percentage was one of the strongest predictors of winning. In their review, Leard and Doyle describe a study by Haisken-DeNew and Vorell (2008), which found that regular-season fighting was correlated with playoff performance from 1996-97 to 2007-08. However, that study measured fighting success using fan polls of who won each fight, which Leard and Doyle argue is likely biased toward the team that won the game. Looking at size directly, Quinney et al.

(2008) studied the preseason fitness testing of 703 players on one NHL team from 1979 to 2005. Players became taller, heavier, and more powerful over time, but the team's average size was nearly identical in its successful and unsuccessful seasons, and its overall fitness profile was not related to team success. Together, these studies suggest that size and toughness may matter less than is commonly believed, but none of them look at physical players as part of a team's full roster composition.

Two other studies are important for how this project should be designed and interpreted. Lopez, Matthews, and Baumer (2018) used betting market data across roughly a decade of play to measure how random the outcomes are in each of the four major North American leagues. The NHL and MLB were the most random, with the better team winning a typical NHL game only about 57% of the time, compared to 67% in the NBA. In a simulated 16-team playoff with best-of-seven series, the best NHL team won the title only about 19% of the time, compared to about 37% in the NBA and NFL. This means that regular-season strength explains only part of NHL playoff results, which is exactly why it is worth asking what else matters. It also sets realistic expectations, since any effect of roster composition will be competing with a large amount of luck. Finally, Schuckers and Macdonald (2014) showed that the NHL's real-time statistics, including hits, are recorded at very different rates in different arenas. For example, games in New Jersey had about 41% fewer hits recorded than expected, while games in Los Angeles had about 30% more. Since hits are one of the main measures of physical play in this project, this is an important limitation to keep in mind.

Overall, the literature shows that data-driven player types are well established in hockey, and that there is growing interest in connecting roster makeup to team success. However, there is still a clear gap. Chan et al. (2012) linked player types to regular-season points, Säfvenberg et al. (2024) linked them to making the playoffs in a single season without controlling for team strength, and Penner (2025) studied championship rosters in basketball without controls. Research on physical play has focused on fighting, salaries, or one team's fitness, rather than how physical players fit into a full playoff roster. No study has tested whether the mix of player types on a team is associated with how far it goes in the playoffs, across many seasons, after accounting for how good the team was in the regular season. This project fills that gap by building player types that include size and physical play from 21 seasons of data, measuring every playoff team's composition by playoff ice time, and testing whether that composition predicts playoff rounds won beyond regular-season strength.

3. Data

All data for this project came from the NHL's public statistics API, and I collected it in Python in three parts: player regular-season statistics, player playoff ice time, and team-level statistics. The sample covers 21 seasons, from 2005-06 through 2025-26. I started with 2005-06 for two reasons. Individual hits and blocked shots, which are key measures of physical play here, were not recorded until after the 2004-05 lockout, and the rule changes that came out of the lockout changed how the game is played. Chan et al. (2012) also limited their data to post-lockout seasons for this reason. The 2025-26 season is the most recent one with completed playoffs.

For the regular season, I joined four API reports for each season using each player's ID: the summary report (goals, assists, shots, penalty minutes, games played, and position), the realtime report (hits and blocked shots), the time-on-ice report (total, power-play, and penalty-

kill ice time), and the bios report (height and weight). The final dataset has 18,853 player-seasons with no missing values, ranging from 839 skaters in the 48-game 2012-13 season to 1,004 in 2021-22, when COVID-19 protocols forced teams to use more call-ups. Goalies were left out, since their job is completely different from a skater's. Goaltending enters the model at the team level instead, through save percentage.

Two details about the API data matter for this project. First, a traded player appears as one row with his combined totals from every team he played for. For example, Jake Guentzel was traded from Pittsburgh to Carolina in 2023-24 and appears once, listed as "PIT,CAR," with all 67 of his games. This works well here, since a player's type is meant to describe his whole season. Second, height and weight are only available as one listed value per player rather than one per season, so a player's listed weight in 2006 may really be his weight from later in his career. I return to this limitation in the Discussion.

For the playoffs, I pulled every skater's playoff ice time, giving 7,286 player-team-seasons. There are no trades during the playoffs, so each row is one player on one team. Every season had 16 playoff teams except 2019-20, which had 24 because of the bubble's qualifying round. In 51 cases, a player appeared in the playoffs without a regular-season row that same season. These were rookies who debuted in the playoffs (ex: Cale Makar, Colorado, 2019) and injured players who returned after missing the whole regular season (ex: Nikita Kucherov, Tampa Bay, 2021). They make up only 0.24% of playoff ice time, but dropping them would remove players like Kucherov, who led the 2021 playoffs in scoring for the Cup champion, from exactly the deep playoff teams the model learns the most from. Instead, I gave each of them his statistics from his nearest regular season (his previous season, or his next one if he had no previous season), and all 51 were matched. A player's style is usually stable from year to year, although a player returning from an injury may play a little differently.

For each team, I used its regular-season points percentage to measure team strength and calculated its save percentage as one minus goals against divided by shots against. The dependent variable is the number of playoff rounds a team won, from 0 to 4. Since every round from 2005-06 on has been a best-of-seven series, a team needs exactly four wins to win a round, so rounds won equals playoff wins divided by four, rounded down (ex: $4 + 4 + 2 = 10$ wins means 2 rounds won). The exception is 2019-20, where the bubble's best-of-five qualifying round and round-robin games break this formula. For that season, I kept the 16 teams that reached the round of 16, dropped the eight qualifying-round losers, and entered the results by hand. The qualifying-round ice time still counts toward the composition of the teams that advanced, since it reflects the lineups they actually used. As a final check, every playoff year has exactly 15 series ($8 + 4 + 2 + 1$), so the rounds won in each season must add up to 15, with one champion winning 4. Every season passed this check, leaving 336 playoff team-seasons ($21 \text{ seasons} \times 16 \text{ teams}$).

Variable	Mean	SD	Min	Max
Playoff rounds won (0–4)	0.938	1.199	0.000	4.000
Regular-season points %	0.626	0.050	0.500	0.823
Regular-season save %	0.909	0.010	0.876	0.933
F: Top-line scorer (%)	34.9	13.1	0.0	71.8
F: Skilled playmaker (%)	26.6	13.0	0.0	69.6
F: Power forward (%)	15.7	9.8	0.0	56.5

Variable	Mean	SD	Min	Max
F: Defensive/PK (%)	19.7	8.6	0.0	43.0
F: Physical (%)	3.0	3.6	0.0	17.3
D: Offensive (%)	35.5	15.5	0.0	74.6
D: Shutdown/PK (%)	30.0	16.1	0.0	68.6
D: Two-way/depth (%)	23.6	17.1	0.0	86.0
D: Physical (%)	10.8	10.6	0.0	49.3

Table 1: Summary Statistics for the 336 Playoff Team-Seasons

Table 1 shows summary statistics for every variable in the regression, with points percentage and save percentage written as decimals. The composition variables (“F:” and “D:”) are each player type’s share of a team’s playoff ice time at that position, which is explained in the Methodology. The mean of rounds won is 0.938 by construction (15 rounds won divided among 16 teams), and its standard deviation is large because half of all playoff teams win zero rounds while one team each year wins four. Points percentage averages .626, well above .500 since every team in the sample made the playoffs, with a high of .823 from the 2022-23 Bruins’ record 135-point season. Save percentage barely varies, with most teams within about a point of .909. The composition variables vary the most, with some types ranging from 0% to over 70% of a team’s ice time, which is the variation the regression relies on. The exception is physical forwards, who averaged only 3.0% of forward ice time and never more than 17.3%, which matters later in the Results.

4. Methodology

This project has three stages. First, I used k-means clustering to sort forwards and defensemen into player types based on size, physical play, production, and special teams role. Second, I described each playoff team’s roster by how much of its playoff ice time went to each type. Third, I used regression to test whether that composition is associated with how far a team goes in the playoffs, after controlling for its regular-season strength. This cluster-then-regress design follows Chan et al. (2012), but with playoff success as the outcome instead of regular-season points.

Stage 1: Defining Player Types

I clustered forwards and defensemen separately, since the positions have different jobs and clustering them together would mostly just separate forwards from defensemen. Only players with at least 20 regular-season games were used to build the clusters, because players with very small samples can have extreme statistics that do not reflect how they play. Chan et al. (2012) and Säfvenberg et al. (2024) used similar playing-time cutoffs. This left 14,227 player-seasons (9,325 forwards and 4,902 defensemen). Player types were defined using regular-season statistics only, since using playoff statistics would partly measure the outcome I am trying to explain. Penner (2025) also left playoff statistics out when building basketball archetypes.

Each player was described by nine features. Six measure production and physical play: goals, assists, shots, hits, penalty minutes, and blocked shots, all per 60 minutes of ice time. Goals and assists were kept separate so the clusters could tell a shooter from a playmaker, and per-60 rates were used so each feature measures how a player plays rather than how much he plays. Two features measure special teams role: the share of a player’s own ice time spent on the

power play and on the penalty kill. A first-liner who plays 20 minutes with 4 on the power play has a power-play share of 0.20, while a fourth-liner who plays 10 minutes with 3 on the penalty kill has a penalty-kill share of 0.30, correctly marking him as a penalty-kill specialist even though he plays half as much. Total ice time was not a feature, following Chan et al. (2012), Penner (2025), and Säfvenberg et al. (2024), since it already enters the model through the Stage 2 weights.

The ninth feature is size. I originally used height and weight as separate features, but they are highly correlated ($r = 0.72$ for forwards), so size was effectively counted twice. In that version, small stars were pulled into a lower-scoring group because of their size, with Johnny Gaudreau landing there in all 10 of his seasons, including his 115-point season in 2021-22. I instead combined height and weight into a single size feature, the average of the two after standardizing. This did not change the fit of the clusters, and since this project asks whether size matters in the playoffs, it avoids building that hypothesis directly into the player types.

All features were standardized into z-scores within each season and position, which puts every feature on the same scale and compares each player only to players at his position in the same season, accounting for how the league has changed over time. I then capped the z-scores at ± 4 . A few players, mostly enforcers, were extreme outliers (167 players were more than four standard deviations above average in penalty minutes alone), and because k-means squares distances, a handful of them could distort the clusters. The cap affected only about 0.2% of values, enforcers still sit at the top of the scale, and the player profiles in the Results use real statistics.

To choose the number of clusters (k), I ran k-means for every k from 2 to 10 at each position, using 20 random starting points per run to avoid poor solutions. For each k , I recorded the inertia (for the elbow method), the silhouette score, and the Calinski-Harabasz index, which Chan et al. (2012) also used. These results are shown in Figure 1.

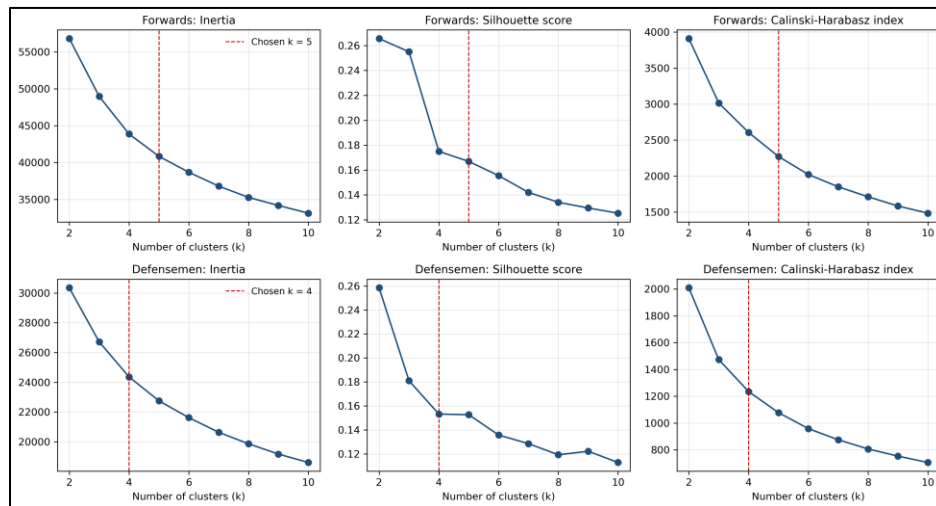


Figure 1: Cluster Quality Measures by Number of Clusters (k)

Neither position shows a sharp elbow, but the gains from each extra cluster shrink noticeably after about four or five clusters. The silhouette score and Calinski-Harabasz index are highest at the smallest values of k , which is typical, and for forwards the silhouette score favored two or three clusters. Those solutions were too broad, though, since with three clusters superstars

were grouped with nearly every middle-six skill player. So, as Chan et al. (2012) did, I used the fit statistics to narrow the range and then chose the number that produced the clearest hockey roles: five for forwards and four for defensemen. For defensemen, a fifth cluster did not improve the silhouette score (0.153 for both) and only split the offensive defensemen into a bigger and a smaller group. Finally, every player-season was assigned to its nearest cluster, including the 4,626 below the 20-game cutoff, so call-ups who played in the playoffs still count toward their team's composition.

Stage 2: Measuring Roster Composition

For each of the 336 playoff team-seasons, I measured composition as the share of the team's playoff ice time that went to each player type, separately for forwards and defensemen, so each position's shares add up to 100%. I weighted by ice time rather than games played because a fourth-liner who plays 8 minutes a night should not count the same as a first-liner who plays 20, and Chan et al. (2012) weighted by ice time as well. As a first look, I compared the average composition of teams grouped by how far they went in the playoffs.

Stage 3: Regression Model

The main model is estimated with OLS:

$$RoundsWon_{it} = \beta_0 + \sum_k \beta_k Share_{k,it} + \beta_8 PointsPct_{it} + \beta_9 SavePct_{it} + \gamma_t + \varepsilon_{it}$$

Here, *RoundsWon* is the number of playoff rounds won (0–4) by team *i* in season *t*, the *Share* terms are the seven composition variables (four forward types and three defense types), *PointsPct* and *SavePct* are regular-season points percentage and save percentage, and γ_t represents season fixed effects. The null hypothesis is that all seven composition coefficients equal zero, meaning roster composition is not associated with playoff success once regular-season strength, goaltending, and season are accounted for.

Because each position's shares add up to 100%, one type at each position has to be left out as the baseline. I used the depth role at each position, Defensive/PK forwards and Two-way/depth defensemen, so each coefficient measures the change in rounds won when 10 percentage points of ice time shift from the depth players to that type. The shares are scaled so one unit equals 10 percentage points, and points percentage and save percentage are scaled so one unit equals one percentage point.

Points percentage controls for team strength. Better teams tend to have more top-line talent and go further in the playoffs, so without this control, a player type could look important simply because good teams have more of them. It also sharpens the question, since the model asks which roster makeups do better or worse in the playoffs than their regular-season record would predict. Save percentage controls for goaltending, since Chan et al. (2012) found that goalies contribute the most to team performance and Leard and Doyle (2011) found save percentage to be a strong predictor of winning. Season fixed effects account for changes in the league over time, and since each season's rounds won add up to 15, they also mean each team is compared to the other teams in the same playoffs. I deliberately left out power-play percentage, penalty-kill percentage, and goal differential. These predict success, but they are partly products of the roster itself. A team with more offensive defensemen will likely have a better power play because of them, so controlling for power-play percentage would absorb part of the effect this project is trying to measure.

The OLS model uses heteroskedasticity-robust (HC1) standard errors, and the null hypothesis is tested with a joint F-test of the seven composition coefficients. Since rounds won is really a set of ordered categories, I also estimated the same model as an ordered logit to check that the results do not depend on how the outcome is modeled. I then ran two robustness checks. The first drops the three unusual seasons: the 48-game 2012-13 season, the 2019-20 bubble playoffs, and the 56-game 2020-21 season. The second addresses the possibility that teams on long playoff runs lean harder on their best players, which would mean playoff ice time partly reflects success. Here, I replaced each player's actual playoff ice time with his expected ice time, equal to his regular-season ice time per game times his playoff games played, which keeps the same players and games but removes any change in usage during the playoffs. Even with these checks, this is an observational study, so the results describe associations, not proof that a roster makeup causes playoff success.

5. Results

Player Types

Tables 2 and 3 show the five forward types and four defense types, with each type's average statistics and its most frequent members. Every type matches a role that coaches and fans would recognize. Top-line scorers get the most power-play time and score the most, and Alex Ovechkin and Sidney Crosby fall into this type in all 21 of their seasons. Skilled playmakers are the smallest forwards and produce mostly through assists. Power forwards are bigger, score at a moderate rate, and hit more than twice as often as the two skilled types. Defensive/PK forwards spend the most time on the penalty kill and block the most shots, while Physical forwards are the heaviest group and have by far the most hits and penalty minutes. On defense, Offensive defensemen lead in power-play time and assists, Shutdown/PK defensemen lead in penalty-kill time and blocked shots, Physical defensemen are the biggest players in the sample and lead in hits and penalty minutes, and Two-way/depth defensemen are close to average on almost everything. These four defense types are nearly identical to the ones found by Chan et al. (2012), even though I found them independently, with different features and 21 seasons of data instead of 5. The types are also stable from year to year. Erik Karlsson is an Offensive defenseman in all 16 of his seasons, Radko Gudas is a Physical defenseman in all 14 of his, and Tom Wilson starts his career as a Physical forward before moving into the more offensive types as his scoring increased.

	Top-line scorer	Skilled playmaker	Power forward	Defensive/PK	Physical
Player-seasons (% of F)	2,020 (21.7%)	2,487 (26.7%)	1,933 (20.7%)	2,173 (23.3%)	712 (7.6%)
Height (in) / Weight (lb)	73.4 / 205	71.6 / 191	73.9 / 208	72.7 / 198	74.3 / 217
PP share (%)	17.0	13.7	7.2	2.5	1.6
PK share (%)	3.0	4.1	3.9	14.2	2.4
Goals/60	1.22	0.77	0.67	0.49	0.42
Assists/60	1.49	1.27	0.80	0.69	0.55
Shots/60	9.39	6.88	7.06	5.60	5.57
Hits/60	3.23	2.76	6.80	6.25	14.92
PIM/60	1.79	1.49	2.53	2.27	12.69
Blocks/60	1.36	1.53	1.78	2.77	2.11

	Top-line scorer	Skilled playmaker	Power forward	Defensive/PK	Physical
Most frequent members (seasons)	Ovechkin (21), Crosby (21), Malkin (20)	Stastny (17), Krejci (15), M. Johansson (15)	J. Staal (13), Foligno (12), Backes (11)	Cizikas (14), N. Thompson (14), Glendening (13)	Reaves (16), M. Martin (15), Neil (12)

Table 2: Forward Player Types

	Offensive	Shutdown/PK	Two-way/depth	Physical
Player-seasons (% of D)	1,234 (25.2%)	1,470 (30.0%)	1,417 (28.9%)	781 (15.9%)
Height (in) / Weight (lb)	73.5 / 205	74.4 / 210	72.9 / 200	75.4 / 220
PP share (%)	13.7	1.9	6.3	1.2
PK share (%)	6.6	12.9	6.5	9.8
Goals/60	0.37	0.13	0.17	0.13
Assists/60	1.12	0.53	0.68	0.48
Shots/60	5.67	3.20	4.00	3.54
Hits/60	2.88	4.19	3.29	8.19
PIM/60	1.57	1.65	1.57	4.12
Blocks/60	3.64	5.41	3.87	4.78
Most frequent members (seasons)	Burns (20), Letang (17), Karlsson (16)	Tanev (14), M. Staal (14), Gorges (13)	Orlov (12), Sekera (11), Hamhuis (11)	Schenn (17), Dillon (14), Gudus (14)

Table 3: Defense Player Types

Roster Composition by Playoff Finish

Figure 2 shows the average composition of playoff teams grouped by how far they went. The clearest pattern is that the share of forward ice time given to Top-line scorers rises steadily with playoff success, from 33.0% for first-round losers to 40.5% for Cup winners. Offensive defensemen follow the same trend (33.5% to 40.9%), while Two-way/depth defensemen fall (25.1% to 19.2%). Physical forwards get slightly less ice time on deeper teams (3.5% down to 2.3%), Power forwards get their lowest share on Cup winners (12.0%), and the other types show no clear trend. Interestingly, regular-season points percentage barely separates the groups, rising only from .613 for first-round losers to .658 for Cup winners, with Final losers (.617) below second-round losers. Before any controls, this first look points away from the idea that size and physicality win in the playoffs, but it does not account for regular-season strength, and the last two groups only contain 21 teams each.

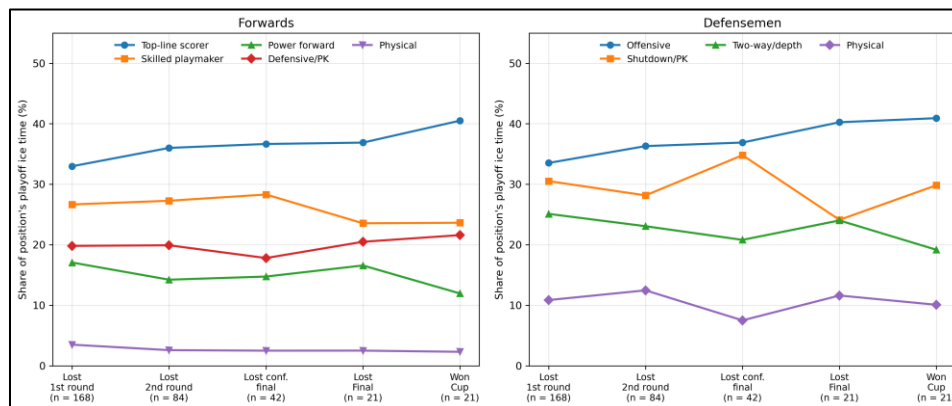


Figure 2: Average Roster Composition by Playoff Finish

Regression Results

Table 4 shows the main regression results. Taken together, the seven composition variables are not statistically significant (joint F-test, $p = 0.152$), so I cannot reject the null hypothesis that roster composition as a whole is unrelated to playoff success. The model explains 9.1% of the variation in rounds won, which is low but not surprising, since a large part of what decides the playoffs is luck that no model can capture. Regular-season points percentage is significant, with 10 more points of points percentage (ex: going from .600 to .700) associated with about 0.46 more rounds won. Regular-season save percentage has no effect once points percentage is in the model, likely because good goaltending is already reflected in a team's points.

Variable	OLS coef. (SE)	OLS 95% CI	Ordered logit OR	OR p-value
F: Top-line scorer	0.054 (0.098)	[-0.139, 0.246]	1.151	0.373
F: Skilled playmaker	0.014 (0.089)	[-0.161, 0.189]	1.119	0.434
F: Power forward	-0.078 (0.092)	[-0.258, 0.102]	0.903	0.491
F: Physical	-0.336* (0.194)	[-0.717, 0.045]	0.482**	0.029
D: Offensive	0.101** (0.048)	[0.006, 0.195]	1.193**	0.026
D: Shutdown/PK	0.036 (0.051)	[-0.064, 0.135]	1.088	0.291
D: Physical	-0.019 (0.070)	[-0.155, 0.118]	0.986	0.901
Regular-season points %	0.046** (0.018)	[0.010, 0.082]	1.091***	0.001
Regular-season save %	0.004 (0.101)	[-0.193, 0.201]	1.047	0.766
Season effects	Yes		Yes	
Observations	336		336	
R-squared	0.091			
Joint test, composition (p)	0.152			

Table 4: Main Regression Results

*Note: Dependent variable is playoff rounds won (0–4). Composition variables are each type's share of the team's playoff ice time at that position, in 10-percentage-point units; the baseline types are Defensive/PK forwards and Two-way/depth defensemen. Points % and save % are in percentage points. OLS uses heteroskedasticity-robust (HC1) standard errors. OR = odds ratio. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.*

Among the individual composition variables, Offensive defensemen stand out. Shifting 10 percentage points of defense ice time from Two-way/depth defensemen to Offensive defensemen is associated with about 0.10 more rounds won ($p = 0.037$). Put in more realistic terms, a one-standard-deviation increase in Offensive defensemen's share (15.5 points) is associated with about 0.16 more rounds won, which is roughly 70% of the gain from a one-standard-deviation improvement in regular-season points percentage (about 0.23 rounds). Physical forwards point in the opposite direction. Since physical forwards average only 3.0% of forward ice time, a 10-point shift is unrealistic, but a one-standard-deviation increase (3.6 points) is associated with about 0.12 fewer rounds won. This effect is only marginally significant in OLS ($p = 0.084$), and its confidence interval is wide because physical forwards get so little ice time that there is not much variation to learn from. The Top-line scorer pattern from Figure 2 disappears completely once points percentage is controlled for ($p = 0.584$), which means that it

mostly reflected better teams simply having more top-line talent. None of the other types have a detectable effect.

The ordered logit tells the same story. Every variable has the same sign as in OLS, and the same variables stand out. An odds ratio of 1.19 for Offensive defensemen means that each 10-point shift toward them is associated with about 19% higher odds of reaching a later round ($p = 0.026$). For Physical forwards, the odds ratio of 0.48 ($p = 0.029$) is significant in this model, even though it was only marginal in OLS.

Robustness Checks

Table 5 and Figure 3 compare the main results to the two robustness checks. Dropping the 2012-13, 2019-20, and 2020-21 seasons leaves 288 team-seasons, and both key results keep their direction and size and stay significant in the ordered logit, with p-values rising slightly as expected with 14% less data. With expected ice time, the shares were almost identical to the originals (correlations of 0.978 to 0.994), so coaches use their players in the playoffs much like they did in the regular season. Every playoff team shifted about one percentage point of ice time toward Top-line scorers and Offensive defensemen, but this shift did not grow with playoff depth (+0.84 points for first-round losers and +1.10 for Cup winners, with no steady trend in between), so deep playoff runs were not driving the ice-time weights. In this version, the results became slightly stronger, with Offensive defensemen significant in both models and Physical forwards significant in the ordered logit and just outside the cutoff in OLS ($p = 0.060$).

Variable	Main OLS	Main OR	Drop seasons OLS	Drop seasons OR	Exp. TOI OLS	Exp. TOI OR
F: Top-line scorer	0.054	1.151	0.110	1.220	0.102	1.266
F: Skilled playmaker	0.014	1.119	0.042	1.164	0.056	1.214
F: Power forward	-0.078	0.903	-0.055	0.936	-0.053	0.939
F: Physical	-0.336*	0.482**	-0.305	0.457**	-0.346*	0.473**
D: Offensive	0.101**	1.193**	0.088*	1.183**	0.115**	1.222**
D: Shutdown/PK	0.036	1.088	-0.006	1.005	0.035	1.087
D: Physical	-0.019	0.986	-0.079	0.912	-0.013	1.001
Points %	0.046**	1.091***	0.043**	1.097***	0.043**	1.086***
Save %	0.004	1.047	-0.054	0.905	0.015	1.073
Observations	336	336	288	288	336	336
Joint test (p)	0.152		0.170		0.065	

Table 5: Robustness Checks

*Note: Dependent variable is playoff rounds won (0–4). Each pair of columns re-estimates the main model: the main specification; excluding the shortened 2012-13 and 2020-21 seasons and the 2019-20 bubble playoffs; and composition weighted by expected ice time (regular-season ice time per game $\times$ playoff games played) instead of actual playoff ice time. All models include season effects; OLS uses HC1 robust standard errors. Baselines and units as in Table 4. OR = odds ratio. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.*

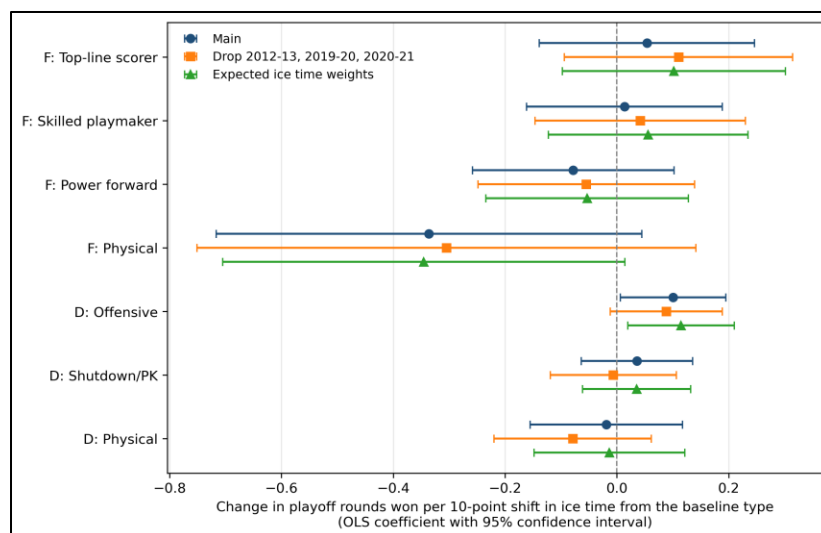


Figure 3: Composition Coefficients Across Model Specifications

Note: Points show OLS coefficients and lines show 95% confidence intervals (HCl robust standard errors) for each type's share of playoff ice time, in 10-percentage-point units, relative to the baseline types. All models control for regular-season points %, save %, and season effects. Intervals that cross the dashed line at zero are not statistically significant at the 5% level.

Across all six models, Offensive defensemen are positive every time and significant at the 5% level in five of them, Physical forwards are negative every time and significant in all three ordered logits, and points percentage is significant in all six. No other composition variable is ever significant, and the composition variables are never jointly significant at the 5% level. In Figure 3, Offensive defensemen are the only type whose confidence intervals sit to the right of zero.

6. Discussion

The biggest takeaway from this project is how hard the playoffs are to predict. Even with regular-season strength, goaltending, season, and seven composition variables, the model explains only about 9% of the variation in playoff rounds won, and regular-season points percentage itself has only a modest effect. This lines up with Lopez et al. (2018), who found that the NHL is one of the most random leagues in North American sports, and it explains why the composition variables are not jointly significant. With this much luck involved, even real effects are hard to separate from noise in 336 team-seasons, which is why the two patterns that held up across all six models are worth taking seriously, even though they are modest.

The first pattern is that teams giving more defense ice time to offensive, puck-moving defensemen go further in the playoffs than their regular-season record would predict. This extends Säfvenberg et al. (2024), who found that playoff teams had more offensive defensemen, by showing that this type is also connected to how deep a team goes once it gets there. One possible explanation is that playoff games are tighter and more heavily checked, so defensemen who can move the puck out of their own zone and create offense from the back end become more valuable when chances are harder to come by. This project cannot confirm why the relationship exists, but it is consistent with a style of play built on skill rather than size.

The second pattern is that teams giving more ice time to physical forwards tend to have shorter playoff runs. This goes against the conventional wisdom about playoff hockey, but it fits

the existing research. Chan et al. (2012) found that physical forwards did not contribute to regular-season points, Leard and Doyle (2011) found that winning fights does not help teams win games, and Quinney et al. (2008) found that a team's size was not related to its success. This project shows that the same holds in the playoffs. Power forwards and physical defensemen, the other types defined by size and physical play, showed no effect either way, so there is no evidence here that size or physicality helps teams in the playoffs. One reason physical forwards may hurt is that every minute they play is taken away from a more skilled player, and their high penalty rates can give opponents more power plays. Still, physical forwards average only 3% of forward ice time, so this effect is uncertain and should be treated as some evidence rather than a firm conclusion.

It is also worth noting what did not hold up. Cup winners gave noticeably more ice time to Top-line scorers, but this disappeared once regular-season strength was controlled for, suggesting that the value of top-line talent is already reflected in a team's regular-season record. For teams building a roster, these results suggest being cautious about adding physical forwards for the playoffs and placing more value on puck-moving defensemen, although the effects are small compared to the role of luck.

There are several limitations to keep in mind. First, the player types describe how a player plays, not how good he is, so players near the edge of two types can move between them. Johnny Gaudreau, a small star, is a Skilled playmaker in 8 of his 10 seasons and a Top-line scorer in the other 2, and Jaccob Slavin, an elite defensive defenseman who rarely hits or takes penalties, is split between Shutdown/PK and Two-way/depth. Second, Schuckers and Macdonald (2014) found that some arenas record far more hits than others, which adds noise to one of the main measures of physical play and could blur the physical types. Third, height and weight are only available as one listed value per player. Fourth, the 51 players who missed the regular season were given statistics from their nearest season, even though a player returning from an injury may play differently. Fifth, the model does not account for the strength of each team's playoff opponents. Finally, this is an observational study, so the results show associations, not proof that a certain roster makeup causes playoff success. Future work could adjust hits for arena effects using the approach of Schuckers and Macdonald (2014) or by only using road games, estimate the model at the series level to include opponent strength, and test fuzzy clustering like Säfvenberg et al. (2024), which allows players to belong to more than one type.

Overall, this project set out to find the ideal makeup of a Stanley Cup roster. The answer is that playoff success depends mostly on regular-season strength and a large amount of luck. Beyond that, the most consistent signal is that the teams that go furthest give more of their ice time to offensive, puck-moving defensemen and less of it to physical forwards. According to this data, the ideal playoff roster is not built around size and toughness, but around skill.

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